# Space Instrumentation Laboratory (SPA617) Record


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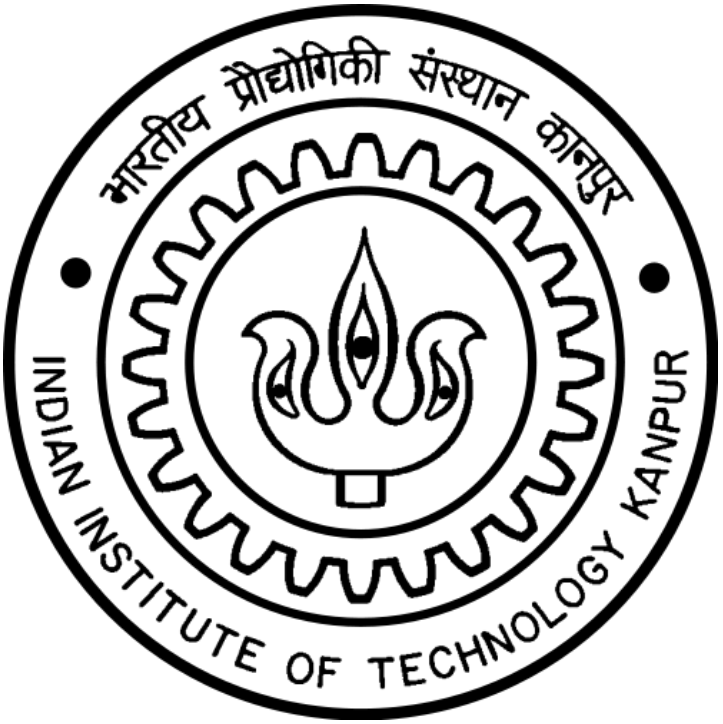
**Semester:** 1st (odd)

**Department:** SPASE

**Institution:** IIT Kanpur


**Academic Year:** 2026-2027

**Course Instructor:** Prof. Mugundhan V



# RADIO ASTRONOMY LABORATORY

## Experiment 1: Antenna Characterization

### 1. Aim

To characterize the given antenna by measuring its beam pattern in both co-polarization and cross-polarization configurations.

### 2. Objectives

**(a)** To determine the antenna operating frequency and return loss from VSWR measurements.

**(b)** To measure the effect of distance on the power received by the antenna.

**(c)** To measure the far-field power pattern of the given antenna with respect to angle in both co-polarization and cross-polarization configurations.

**(d)** To determine the half-power beamwidth and estimate the directivity of the antenna

### 3. Components Required:

- Monopole and dipole antennas
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- 50 Ω termination
- Antenna support stand
- Coaxial cables and suitable RF adapters

### 3. Theory

#### 3.1 Antenna

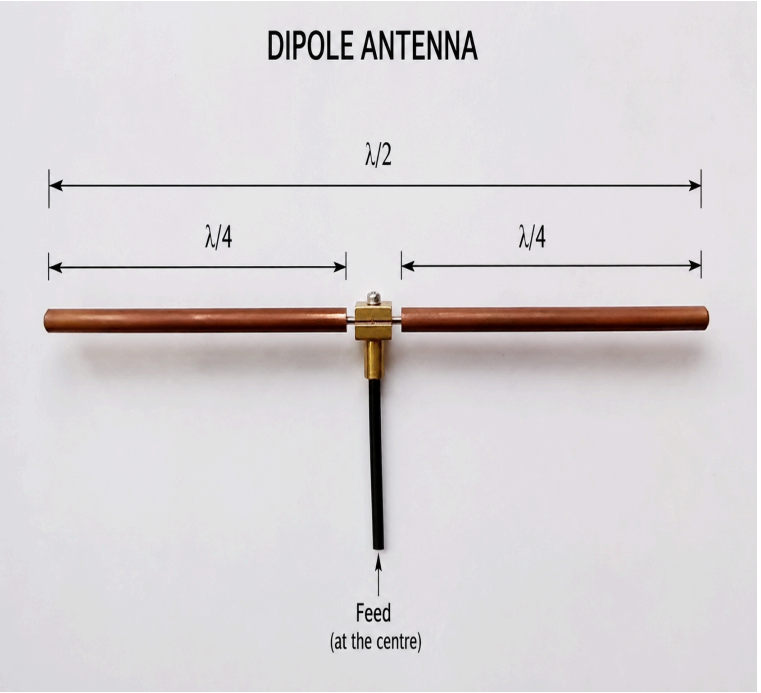
An antenna is a transducer that converts electrical power into electromagnetic waves during transmission and electromagnetic waves into electrical signals during reception.

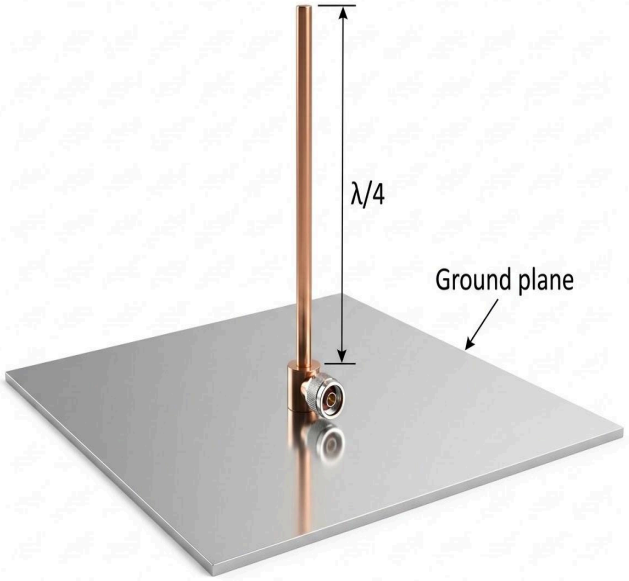
A dipole antenna consists of two approximately equal conducting arms fed at the centre. Its total length is approximately:

$$L \approx \frac{\lambda}{2}$$

A monopole antenna is approximately half of a dipole and is mounted above a conducting ground plane. Its length is approximately:

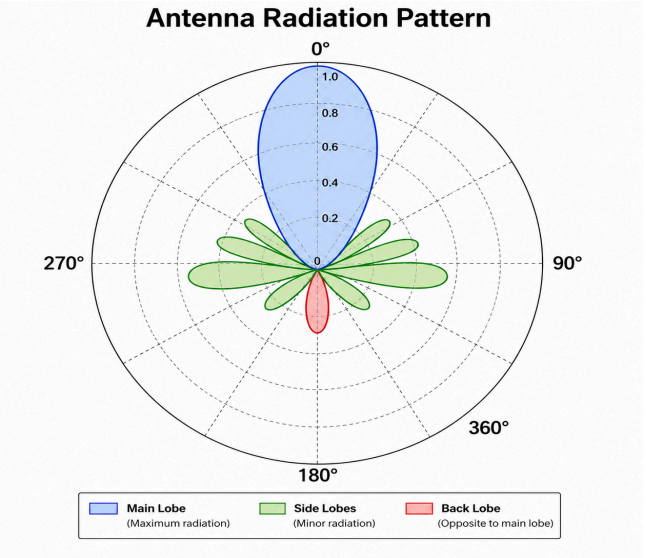
$$L \approx \frac{\lambda}{4}$$






The ground plane acts approximately as an electromagnetic mirror, making the monopole behave similarly to a full dipole.

**Figures: Dipole and monopole Antenna**

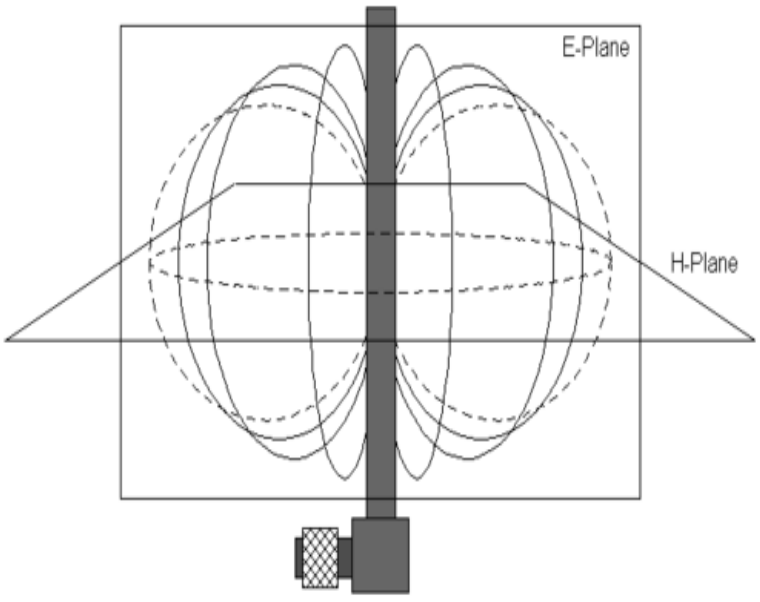



### 3.2 Radiation Pattern

The **radiation pattern** represents how the energy radiated by an antenna is distributed with direction.

It generally contains:

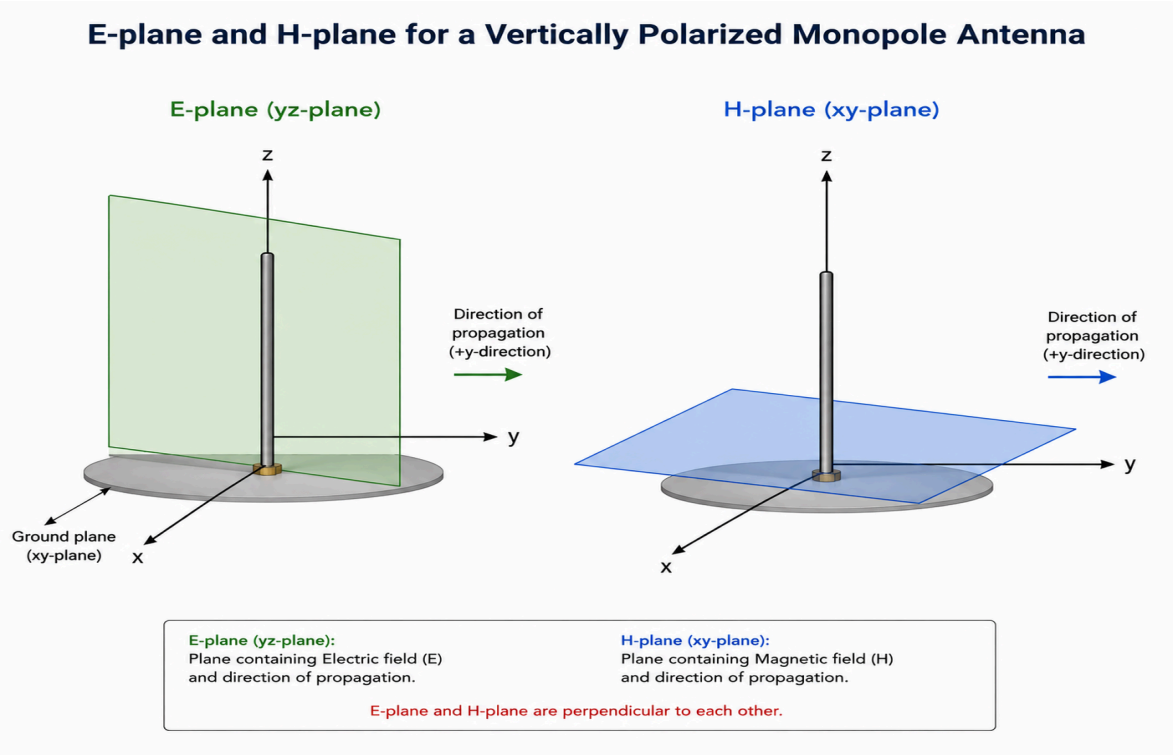
- **Main lobe:** region of maximum radiation.
- **Side lobes:** smaller lobes away from the main lobe.
- **Back lobe:** lobe in the direction opposite to the main lobe.




The E-plane and H-plane are used to measure the angular beam pattern.

The **E-plane** contains the electric-field vector and propagation direction, while the **H-plane** contains the magnetic-field vector and propagation direction. The two planes are perpendicular to each other.

**Figure:** E-plane and H-plane for a vertically polarized monopole Antenna
(Credit: Wikipedia)




## 3.4 Half-Power Beamwidth

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power falls to half of its maximum value.

These points correspond to the −3 dB points.

A small HPBW corresponds to a narrow and more focused beam, while a large HPBW corresponds to a broader beam.

## 3.5 Directivity

Directivity describes how concentrated the radiation of an antenna is in a particular direction compared with an isotropic antenna.

The directivity is related to the beam solid angle $\Omega_A$ by

$$D=\frac{4\pi}{\Omega_A}$$

For a practical antenna, the beam solid angle can be approximately determined using the half-power beamwidths of the E-plane and H-plane:

$$D\approx\frac{4\pi}{\theta_E\theta_H}$$

where $\theta_E$ and $\theta_H$ are the E-plane and H-plane HPBWs, respectively, and must be expressed in radians.

## 4. Experimental Setup

For the antenna measurements, the transmitting antenna is connected to the RF signal generator and the receiving antenna is connected to the spectrum analyser. The antennas are maintained at the same height and aligned appropriately for the measurement. The monopole antenna available for the experiment had a length of approximately **5 cm**. Therefore, we selected an operating frequency of approximately **1.4 GHz**, for which quarter-wavelength is close to the length of our antenna.

## 5. Observation

## 5.1 Effect of Distance

In this part of the experiment, we studied how the received power changes, when the distance between the transmitting and receiving antennas is changed.

As an electromagnetic wave propagates through free space, its power density decreases with increasing distance from the transmitting antenna. In the far-field region, or in the expected free-space the received power approximately follows the inverse-square law,

$$P_r \propto \frac{1}{R^2}$$

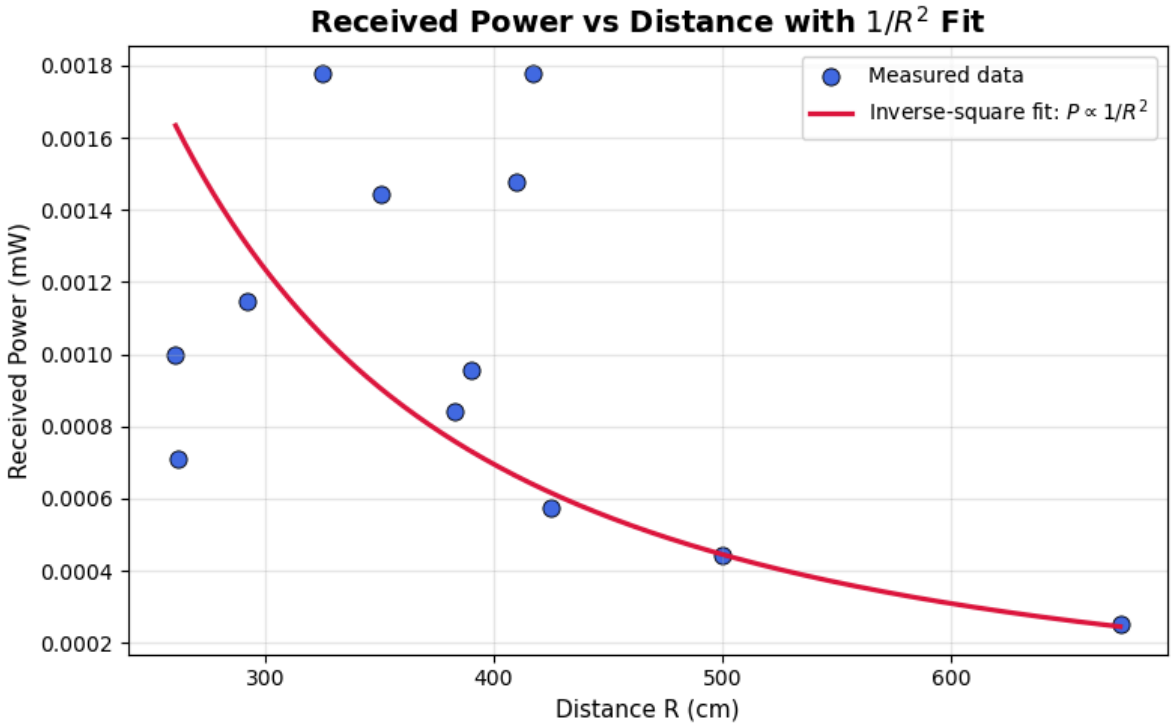
where Pr is the received power and r is the separation between the transmitting and receiving antennas.

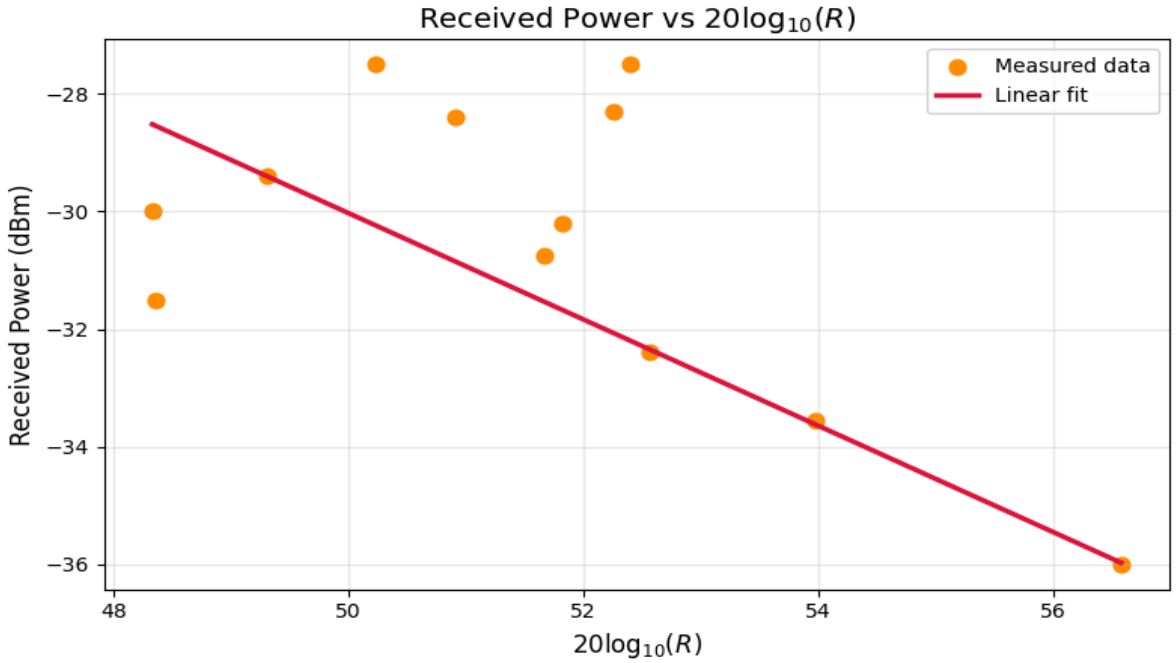
We then changed the separation distance (R) between the two antennas. For each distance, we observed the received signal and recorded its **received power in dBm**. A representative value was therefore recorded for each distance.

**Table 1:** Received power and corresponding 20 log10 R for different antenna separations

| Distance (R) (cm) | Received Power (dBm) | Received Power (mW) | $20log_{10}(R)$ |
|---|---|---|---|
| 261 | −30.00 | 0.001 | 48.333 |
| 262 | −31.50 | 0.000708 | 48.366 |
| 292 | −29.40 | 0.001148 | 49.308 |
| 325 | −27.50 | 0.001778 | 50.238 |
| 351 | −28.40 | 0.001445 | 50.906 |
| 383 | −30.75 | 0.000841 | 51.664 |

| | | | |
|---|---|---|---|
| 390 | −30.20 | 0.000955 | 51.821 |
| 410 | −28.30 | 0.001479 | 52.256 |
| 417 | −27.50 | 0.001778 | 52.403 |
| 425 | −31.40 | 0.000724 | 52.568 |
| 500 | −33.55 | 0.000442 | 53.979 |
| 674 | −36.00 | 0.000251 | 56.573 |



These graphs may seem a little bit diverting from the actual theoretical fit, but the possible reason for that is  at a fixed distance, the power level was not perfectly constant; the peak amplitude fluctuated continuously. This may be due to reflections from nearby walls, tables, and other objects in the laboratory environment, as well as other measurement effects.

## 5.2 Beam pattern for co-polarization measurement

In this part, the directional response of the receiving horn antenna was measured in the co-polarized configuration. Co-polarized means that electric fields of the transmitting and receiving antennas were aligned in the same plane. Both antennas were maintained at the same height. The receiving antenna was first rotated from $0^o\ to\ 360^o$, and the received power was recorded at suitable angular intervals, and then we repeated the same procedure for the H-plane.

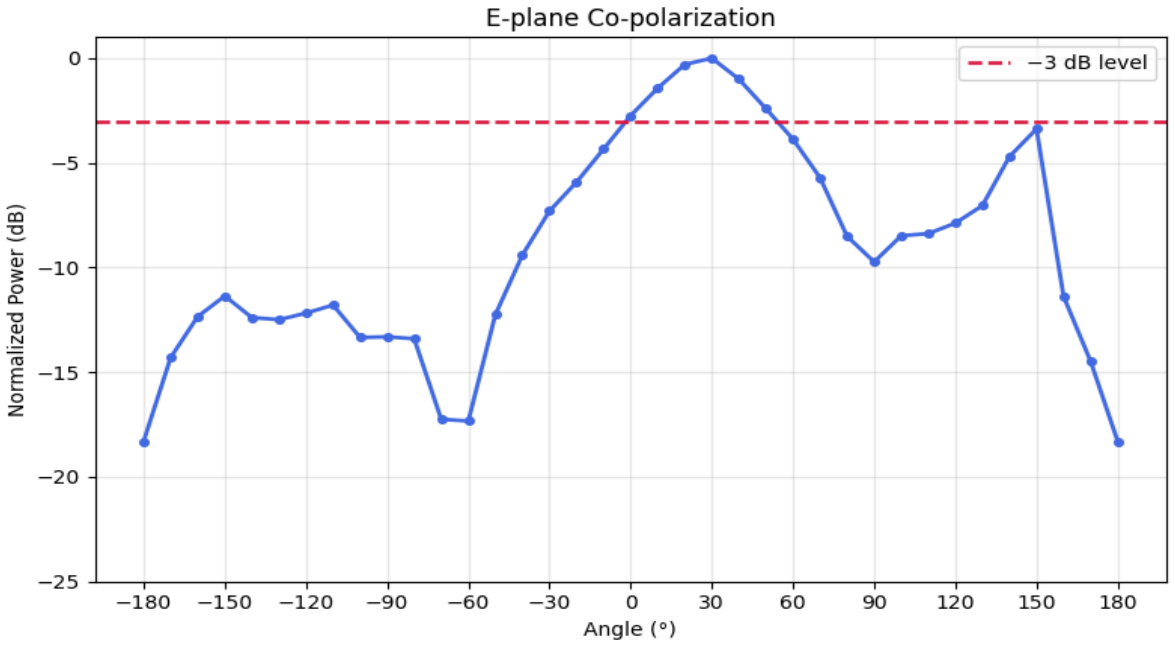
**Table 2: Measured Received Power as a Function of Angle — Co-polarization**

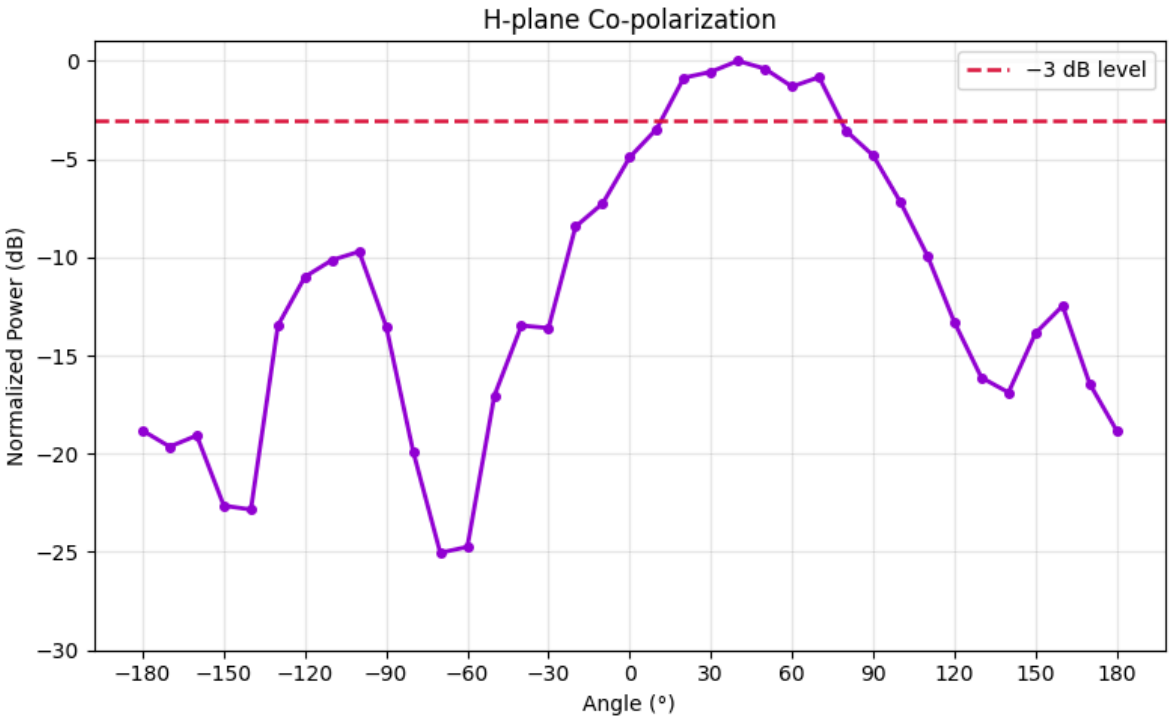
| Angle (°) | Power in E (dBm) | Power in H (dBm) |
|---|---|---|
| −180 | −49.02 | −48.98 |
| −170 | −45.02 | −49.80 |
| −160 | −43.06 | −49.23 |
| −150 | −42.08 | −52.80 |
| −140 | −43.11 | −53.00 |
| −130 | −43.21 | −43.63 |
| −120 | −42.90 | −41.15 |
| −110 | −42.51 | −40.30 |
| −100 | −44.06 | −39.87 |
| −90 | −44.03 | −43.67 |
| −80 | −44.12 | −50.05 |
| −70 | −47.97 | −55.20 |
| −60 | −48.05 | −54.90 |
| −50 | −42.97 | −47.20 |
| −40 | −40.12 | −43.63 |
| −30 | −38.02 | −43.76 |
| −20 | −36.64 | −38.60 |
| −10 | −35.06 | −37.42 |

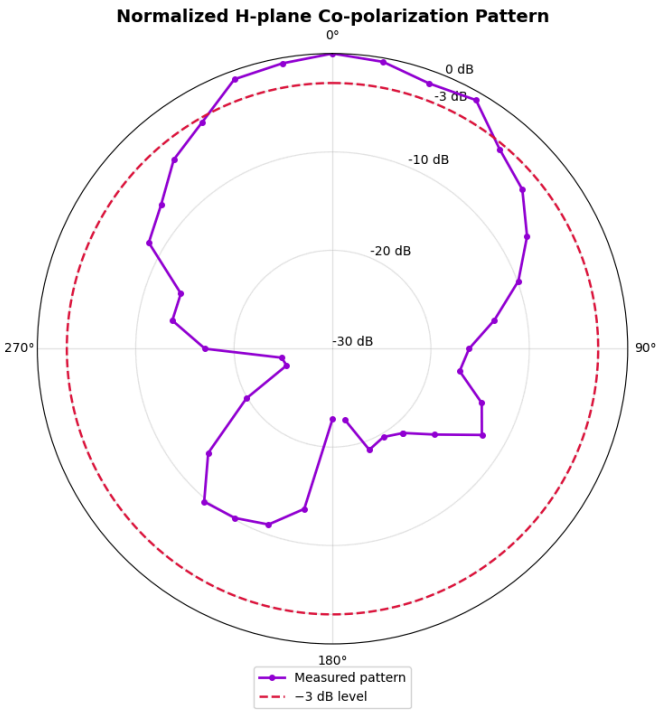
| | | |
|---|---|---|
| 0 | −33.46 | −35.07 |
| 10 | −32.15 | −33.62 |
| 20 | −31.02 | −31.02 |
| 30 | −30.72 | −30.72 |
| 40 | −31.72 | −30.16 |
| 50 | −33.13 | −30.56 |
| 60 | −34.59 | −31.47 |
| 70 | −36.43 | −30.99 |
| 80 | −39.23 | −33.72 |
| 90 | −40.45 | −34.98 |
| 100 | −39.20 | −37.33 |
| 110 | −39.10 | −40.06 |
| 120 | −38.60 | −43.46 |
| 130 | −37.76 | −46.28 |
| 140 | −35.42 | −47.04 |
| 150 | −34.10 | −44.02 |
| 160 | −42.10 | −42.63 |
| 170 | −45.20 | −46.59 |
| 180 | −49.02 | −48.98 |

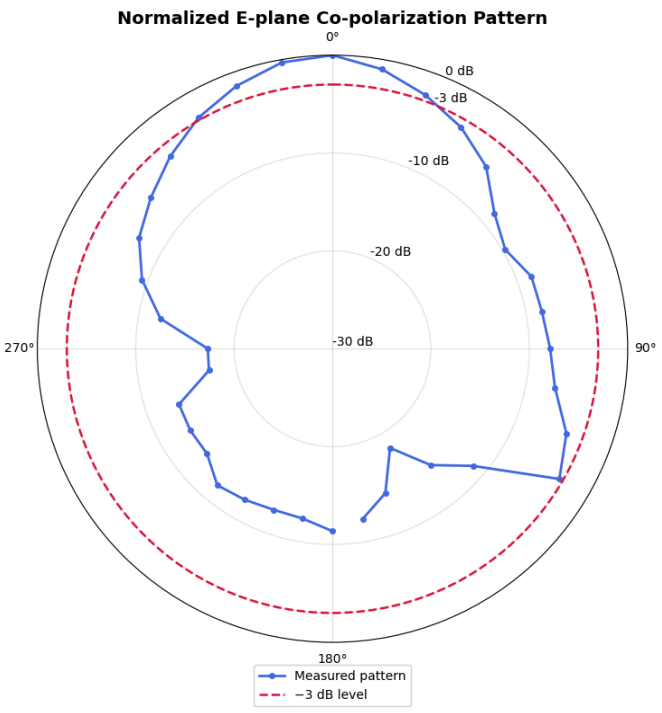
The measured received power was plotted against angle using polar plots. These plots show how the response of the antenna changes with its direction.

**Received power as a function of antenna angle and their patterns:**






The beam patterns obtained for the E- and H-planes clearly show that the antenna response is not confined to a single direction, but it has significant responses in a wide range of angles.

## 5.3 Beam pattern – cross-polarization measurement

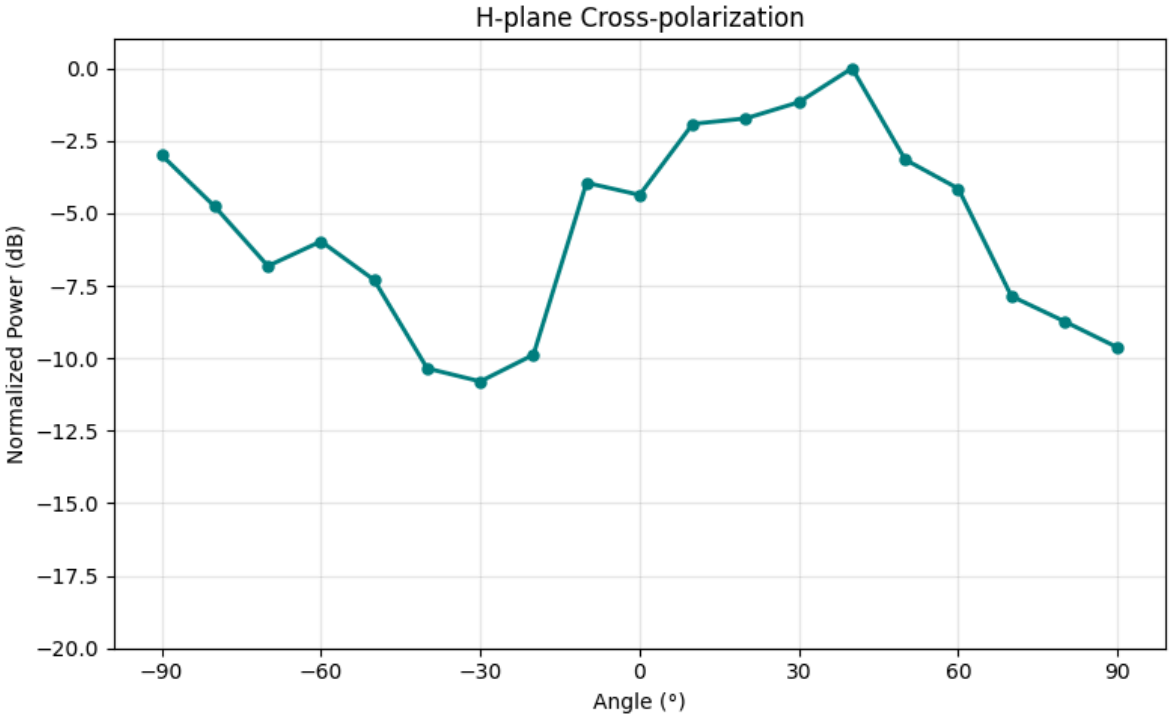
In this case, the transmitting antenna was adjusted so that the electric fields of the transmitting and receiving antennas were perpendicular to each other. This configuration is called cross-polarization.

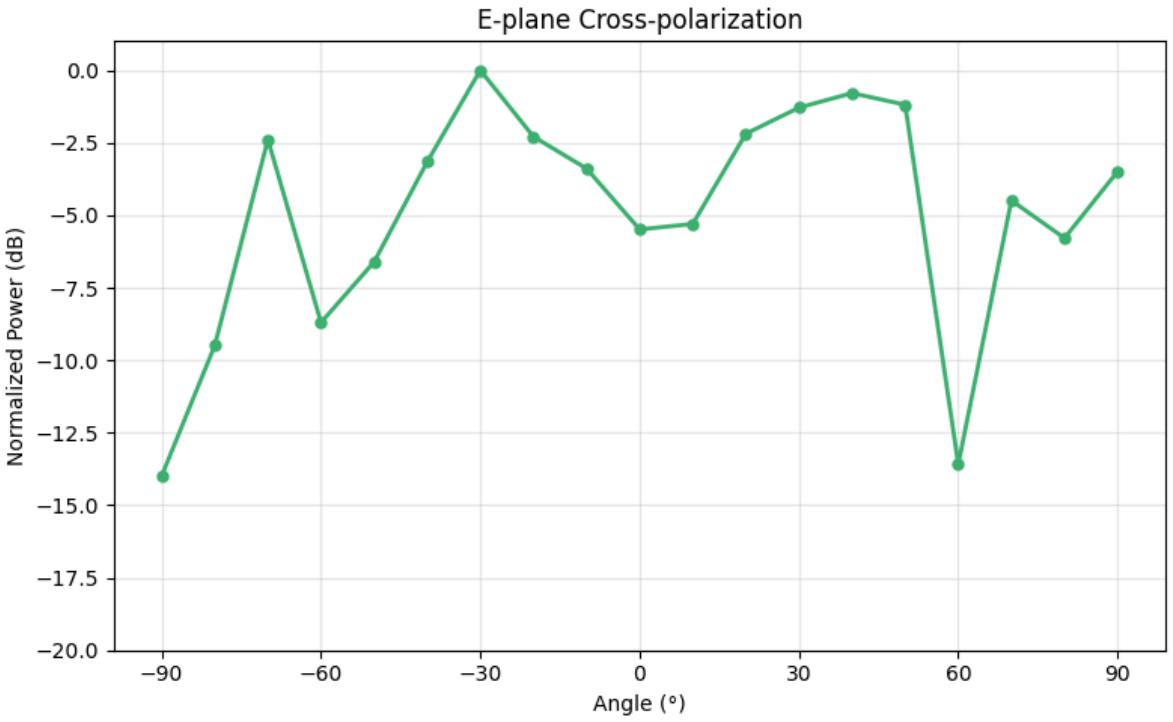
The receiving antenna was rotated through the E-plane and H-plane, and the received power was recorded at each angular position.

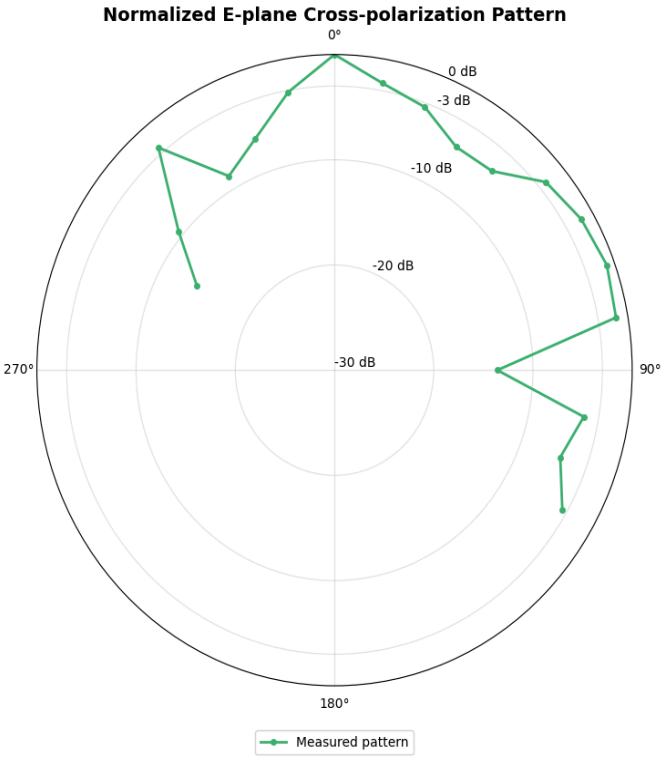
**Table 2:** Measured Received Power as a Function of Angle — Cross-polarization

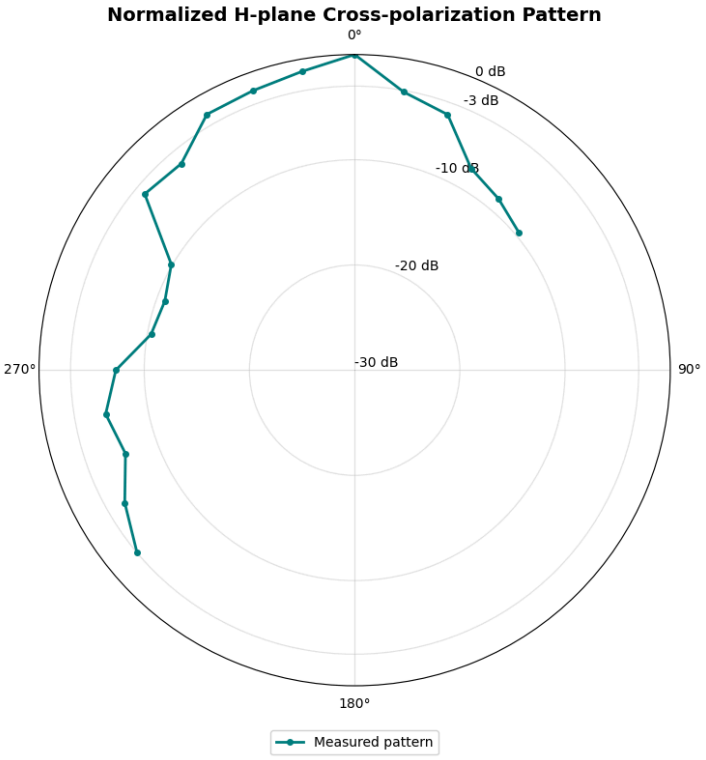
| Angle (°) | Power in E (dBm) | Power in H (dBm) |
|---|---|---|
| −90 | −60.00 | −42.27 |
| −80 | −55.50 | −44.04 |
| −70 | −48.40 | −46.09 |
| −60 | −54.71 | −45.24 |
| −50 | −52.62 | −46.57 |
| −40 | −49.17 | −49.62 |
| −30 | −46.01 | −50.07 |
| −20 | −48.30 | −49.15 |
| −10 | −49.40 | −43.22 |
| 0 | −51.50 | −43.64 |
| 10 | −51.31 | −41.19 |
| 20 | −48.20 | −41.00 |
| 30 | −47.30 | −40.44 |
| 40 | −46.80 | −39.27 |
| 50 | −47.20 | −42.42 |
| 60 | −59.60 | −43.42 |
| 70 | −50.50 | −47.13 |
| 80 | −51.80 | −48.00 |
| 90 | −49.50 | −48.90 |

**The measured values were then plotted as polar beam patterns.**


H-plane Cross-polarization
Normalized Power (dB)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
−90
−60
−30
0
30
60
90
Angle (°)


E-plane Cross-polarization
Normalized Power (dB)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
−20.0
−90
−60
−30
0
30
60
90
Angle (°)




When cross-polarized measurements are compared with the co-polarized measurements, and When the antennas are co-polarized, their polarization directions are aligned, so the receiving antenna generally gives a stronger response. When they are cross-polarized, the polarization directions are perpendicular, so the received response is generally weaker.

Therefore, it helps us understand how strongly the antenna response depends on polarization alignment. For the present measurements, the cross-polarized patterns also show several variations with angle. These variations can be caused by the antenna response as well as reflections and interference from the laboratory surroundings.

## 6. Half-Power Beamwidth and Directivity calculation

The half-power beamwidth (HPBW) and directivity of the antenna were determined from the co-polarized beam patterns. The calculation is performed separately for the E-plane and H-plane.

First, the maximum received power in the beam pattern is identified. If the maximum received power is $P_{max}$, the half-power level in dB is approximately 3 dB below the maximum:

**For the E-plane t**he maximum received power was measured to be:

$$P_{max,E} = -30.72\ dBm$$

Therefore, the half power level should be,

$$P_{-3dB,E} = -30.72\ dBm - 3$$

$$P_{-3dB,E} = -33.72\ dBm$$

Therefore, the two angles at which the beam pattern crosses the −3 dB level are identified. The measured data normally do not fall exactly on the −3 dB value. Therefore, linear interpolation is used to estimate the angle at which the curve actually reaches the −3 dB level.

**First −3 dB crossing**

The first −3 dB crossing occurs between $-10^{o}$ and $0^{o}$At these points the measured power values are −35.06 dBm and -33.46 dBm. Using linear interpolation:

$$\theta_{E,L} = -10 + \frac{-33.46-(-35.06)}{-33.72-(-35.06)}(0 - (-10))$$

$$\theta_{E,L} \approx -1.63^{o}$$

Thus,

$$\theta_{E,L} \approx -1.63^{o}$$

**Second −3 dB crossing**

The second crossing occurs between $50^{\circ}$and $60^{\circ}$, Thus,

$$\theta_{E,RL} = 50 + \frac{-34.59-(-33.13)}{-33.72-(-33.13)}(60 - 50)$$

$$\theta_{E,R} \approx 54.04^{o}$$

Thus,

$$\theta_{E,R} \approx 54.04^{o}$$

Hence, the E-plane half-power beamwidth is the angular separation between these two points:

$$HPBW_{E} = \theta_{E,R} - \theta_{E,L}$$

$$HPBW_E = 54.04^o - 1.63^o$$

$$HPBW_E = 55.67^o$$

For the H-plane, the maximum received power was:

$$P_{max,H} = -30.16\ dBm$$

Therefore:

$$P_{-3dB,H} = -30.16 - 3$$

$$P_{-3dB,H} = -33.16\ dBm$$

**First −3 dB crossing**

The first crossing occurs between 10◦ and 20◦, Using interpolation:

$$\theta_{H,L} = 10 + \frac{-31.02-(-33.62)}{-33.16-(-33.62)}(20 - 10)$$

$$\theta_{H,L} \approx 11.77$$

Therefore,

$$\theta_{H,L} \approx 11.77$$

**Second −3 dB crossing**

From the measured data, the crossing is between 70◦ and 80◦, Therefore:

$$\theta_{H,L} = 70 + \frac{-33.72-(-30.99)}{-33.16-(-30.99)}(80 - 70)$$

Thus,

$$\theta_{H,R} \approx 77.95\circ$$

The H-plane HPBW is:

$$HPBW_H = \theta_{H,R} - \theta_{H,L}$$

$$HPBW_H = 77.95 - 11.77$$

$$HPBW_H \approx 66.18$$

The directivity formula requires the beamwidths in radians.

The conversion is:

$$\theta_{rad} = \theta_{deg} \frac{\pi}{180}$$

For the E-plane:

$$\theta_E = 55.67 \frac{\pi}{180}$$

$$\theta_E \approx 0.9716\ rad$$

For the H-plane:

$$\theta_H = 66.18 \frac{\pi}{180}$$

For the E-plane:

$$\theta_H \approx 1.1551\ rad$$

**Now, Directivity Calculation**

The directivity is estimated using:

$$D \approx \frac{4\pi}{\theta_E \theta_H}$$

Substituting the measured beamwidths:

$$D \approx \frac{4\pi}{(0.9716)(1.1551)} = 11.20$$

Therefore,

$$D \approx 11.20$$

The calculated directivity of approximately 11.20 indicates that the antenna has a measurable preference for certain directions, despite the presence of significant side and back lobes in the measured beam patterns.

## 7. Conclusion

As a final result, the received power varied noticeably with the orientation of the receiving antenna in both the E-plane and H-plane measurements. In the co-polarization measurements, the maximum received power was approximately **−30.72 dBm in the E-plane** and **−30.16 dBm in the H-plane**. In comparison, the cross-polarization measurements generally showed lower received power.

Multiple peaks were observed in the measured patterns for both polarization configurations. This indicates that the antenna response was not limited to a single direction. The additional peaks may be caused by reflections and interference from the surrounding laboratory environment, along with the actual radiation characteristics of the antenna.

The co-polarized configuration generally produced a stronger received signal because the polarization of the incident wave was aligned with the receiving antenna. In the cross-polarized configuration, the electric-field orientations were perpendicular, resulting in a weaker overall response.

A difference was also observed between the E-plane and H-plane patterns. This shows that the measured antenna response depends on both the **orientation of the antenna** and the **polarization of the received wave**. Overall, the measurements demonstrate that antenna orientation and polarization have a significant effect on the received radio-wave power.

And Finally, in this experiment, we understood the basic working of an antenna and how it can be used to transmit and receive electromagnetic waves. We studied the important characteristics of monopole and dipole antennas and observed how antenna performance changes with frequency, distance, and direction.

**The question was asked to me: How do multiple lobes form when the antenna is electrically too long for the frequency?**

Multiple lobes can form because, when the wavelength becomes shorter than the antenna length, the antenna becomes electrically long. The wave travelling along the antenna is reflected at the open ends, and the incident and reflected waves superimpose to form a standing-wave current pattern with several maxima and minima along the antenna. These different current regions radiate with different phases, so their radiation can add constructively in some directions and destructively in others. This produces multiple radiation maxima, or lobes.

## 7. Sources of Error

The following factors may have contributed to deviations between the experimental results and the expected theoretical behaviour:

- **Antenna alignment:** Small changes in the relative orientation or position of the transmitting and receiving antennas can affect the received power.

- **Reflections from surroundings:** Walls, tables, floors, and other objects in the laboratory can reflect electromagnetic waves. These reflected waves can combine with the direct signal and cause fluctuations in the measured power.
- **Signal fluctuations:** The received peak power was observed to fluctuate continuously even when the antenna distance was kept fixed. Recording a single representative value from this fluctuation introduces some measurement uncertainty.
- **Environmental interference:** Electrical equipment, lighting systems, cables, and other electronic devices in the laboratory may introduce unwanted electromagnetic signals.
- **Distance measurement:** Small errors in measuring the separation between the antennas can affect the distance-dependent power measurements.
- **Cable and connector losses:** Losses and imperfect connections in coaxial cables, adapters, and connectors can affect the measured signal level.
- **Non-ideal laboratory environment:** The experiment was performed in a normal indoor environment rather than an ideal free-space or anechoic environment, so the measured radiation pattern and received power may differ from theoretical predictions.

## Disclose Use of AI

AI tools were used during the experiment for the following purposes:

- **Content preparation:** Helped organize and write the laboratory report in a clear and structured format.
- **Study assistance:** Helped understand important antenna concepts such as dipole and monopole antennas, wavelength, impedance matching, radiation patterns, E-plane and H-plane, VSWR, return loss, HPBW, and directivity.
- **Conceptual understanding:** Helped explain the working principles of the experimental setup and the purpose of different components.
- **Data analysis:** Assisted in organizing the experimental measurements and identifying the maximum and relevant values from the recorded data.
- **Data plotting:** Helped prepare Python code for plotting the measured antenna radiation patterns and normalizing the data.
- **Calculations:** Assisted with calculations of HPBW and directivity from the measured radiation-pattern data.